

Automated Sessile-Drop Contact Angle Analyzer

Scientific-Precision Contact-Angle Metrology on a Smartphone

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By

Yennampelly Uday Kiran

Roll No. 22B2509

Under the supervision of Prof. Sudhanshu Mallick



INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

ज्ञानम् परमम् ध्येयम्

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Dissertation Approval Sheet

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Chairperson:

Supervisor: Prof. Sudhanshu Mallick

Examiner 1:

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Yennampelly Uday Kiran
22B2509

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Yennampelly Uday Kiran**

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Abstract

Contact-angle goniometry is the standard technique for quantifying the wettability of solid surfaces, but commercial instruments cost on the order of ₹15 lakh, placing routine surface metrology out of reach of many laboratories. This project developed a complete, low-cost sessile-drop contact-angle analyzer: a back-lit optical bench (LED source, X/Z sample stage, smartphone mount; fabricated cost $\approx$ ₹18,000) driven by a Flutter/Dart mobile application whose image-analysis pipeline was engineered, and — critically — *validated*, to scientific standards.

The pipeline extracts the drop silhouette with sub-pixel edge localisation, detects the substrate baseline through a hierarchy of physically motivated detectors (matte band clustering, specular mirror-symmetry, glossy-surface step detection, and drop/reflection junction-corner location following the practice of commercial instruments), and determines the contact angle by an ensemble of circle, ellipse, rotated-frame polynomial and full Young–Laplace (axisymmetric drop-shape analysis, ADSA) fits with strict validity gating. Every reported angle carries a propagated uncertainty whose dominant term — baseline placement sensitivity, $d\theta/dh = 1/(r \cdot \sin \theta)$ — follows the published error analysis of goniometry, together with ISO 19403-2-style measurement-quality warnings.

Accuracy was established on three independent tiers: (i) a supersampled synthetic harness with exactly known angles, on which the pipeline achieves a mean absolute error of **0.106°** (60–145°) and **0.135°** under noise, stage tilt, defocus blur, small drops and contact-shadow stress; (ii) twelve PFOTES-coated substrate drops referenced against laboratory LB-ADSA fits, agreeing to **1.56° MAE** — within the reference method's own scatter; and (iii) an external published dataset (DropletLab Teflon series) with dispensing needles, glossy substrates and faint reflections, on which targeted baseline research reduced the error from 21.2° to **7.3° MAE** without any regression on the primary rig. The instrument thus delivers laboratory-defensible static contact angles at roughly 1% of commercial cost.

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Chapter 1: Introduction

1.1 Overview

The contact angle θ that a liquid drop makes with a solid substrate is the primary observable of wetting: it encodes the balance of solid–vapour, solid–liquid and liquid–vapour interfacial energies through Young's equation [1], and underpins characterisation of coatings, surface treatments, adhesion, and super-hydrophobic materials. In practice θ is measured optically by the *sessile-drop method*: a small drop is imaged side-on as a silhouette against a diffuse back-light, its profile is extracted, and the tangent at the three-phase contact line is computed.

Commercial goniometers (Krüss DSA series, Biolin Attension Theta, ramé-hart, DataPhysics OCA) reach $\pm 0.1^\circ$ instrument precision, but notably do so with modest $\sim 1\text{--}5$ MP cameras: their accuracy lives in the *optics* — telecentric lenses, homogeneous monochromatic back-lights, levelled precision stages — and in careful software, not in sensor resolution. This observation motivated the present project: a modern smartphone sensor far exceeds the resolution of these instruments' cameras, so a carefully designed optical bench plus rigorous on-device software should approach laboratory-grade results at a small fraction of the cost. The physical instrument (LED back-light, X/Z scissor stage, acrylic frame, smartphone mount) was fabricated for $\approx ₹18,000$ — a $>95\%$ cost reduction.

1.2 Problem Statement

The difficulty of automated sessile-drop analysis is not the fitting mathematics but the *robustness of geometry extraction on real images*. Field images exhibit: (a) dark drops merging optically with dark substrates, defeating gradient edge detectors; (b) interior refraction windows and glare creating spurious internal edges; (c) an ambiguous baseline on reflective or glossy substrates, where the drop's mirror image, its shadow, or the substrate's front edge can each be mistaken for the contact line — published error analysis shows a single pixel of baseline error costs $\approx 0.5\text{--}1^\circ$ below 150° and up to 8° near 180° [9]; (d) dispensing needles piercing the apex; and (e) stage tilt. Naive single-model fitting (e.g. a circle) additionally ignores gravitational flattening at Bond numbers above ≈ 0.1 .

1.3 Objectives and Contributions

The specific contributions of this work are:

1. A fully native Dart image-analysis pipeline — no desktop dependencies — implementing sub-pixel silhouette extraction, a physically-motivated multi-detector baseline hierarchy, four independent angle-extraction methods including a complete Bashforth–Adams ADSA solver, robust ensemble combination, and a propagated uncertainty budget with ISO-style quality flags.
2. A three-tier validation methodology: exact synthetic ground truth (the only way to measure true accuracy, since laboratory references are themselves fits), a laboratory LB-ADSA reference series, and an independent external dataset — with all thresholds locked into an automated regression-test suite so that accuracy cannot silently degrade during development.

3. A metrology-aware capture subsystem exploiting the phone's own sensors: live level/pitch readouts against ISO 19403-2 recommendations, focus and exposure locking, and automatic post-capture exposure quality control.

Chapter 2: Background and Theory

2.1 Contact-Angle Thermodynamics

For a drop at equilibrium on an ideal rigid solid, Young's equation relates the contact angle to the three interfacial tensions [1]:

$$\gamma_{SV} - \gamma_{SL} = \gamma_{LV} \cos \theta \quad (1)$$

Surfaces with $\theta < 90^\circ$ are hydrophilic, $\theta > 90^\circ$ hydrophobic, and $\theta > 150^\circ$ super-hydrophobic. Real surfaces exhibit contact-angle hysteresis between advancing and receding values; the present work measures the static angle of a gently deposited drop, with left and right angles reported separately as a consistency check.

2.2 The Young–Laplace Equation and Drop Shape

The equilibrium meridian profile of an axisymmetric sessile drop satisfies the Young–Laplace equation. In the Bashforth–Adams arc-length parametrisation [2], with lengths scaled by the apex radius of curvature b and φ the tangent angle from the horizontal:

$$dx/ds = \cos \varphi, \quad dz/ds = \sin \varphi, \quad d\varphi/ds = 2 - \beta z - \sin \varphi / x \quad (2)$$

where $\beta = \Delta\rho \cdot g \cdot b^2 / \gamma$ is the shape (Bond) parameter. As $\beta \rightarrow 0$ the solution degenerates to a spherical cap; gravity flattens larger drops. Two consequences matter for metrology. First, only a full Young–Laplace fit is unbiased for large drops. Second, at small β the parameter becomes unidentifiable — the fit degenerates to a circle and surface-tension information is lost; Zuo et al. formalise this with a critical Neumann number below which ADSA-derived tension is unreliable while the *angle* remains valid [8]. The pipeline reports the fitted β and flags this regime rather than over-claiming.

A useful exact bound follows from cap geometry: a profile of height h and contact half-width a satisfies $\theta \geq 2 \cdot \text{atan}(h/a)$, with equality for a spherical cap. This bound is used in the pipeline as a physical gate against degenerate fits (§4.5).

2.3 Contact-Angle Extraction Methods and Their Biases

Four families of methods are standard, each with documented failure modes [6,7,10]:

Circle fit. Exact for $\beta \rightarrow 0$; systematically underestimates θ for gravity-flattened drops. Best for small drops; ISO 19403-2 prescribes it below 20° [12].

Ellipse fit. Accommodates first-order flattening (ISO regime 20 – 110°); over-flexes on near-spherical small drops and is sensitive to non-axisymmetry.

Polynomial/tangent methods. Fit the near-contact profile directly; make no shape assumption but are documented to be sensitive to polynomial order and to the number of contour pixels included [6]; ISO prescribes tangent/Young–Laplace above 110° .

ADSA (Young–Laplace fitting). The physical reference method [3,4]: fit the integrated equation (2) to the observed contour, reading θ where the fitted profile crosses the baseline. Requires axisymmetry, orthogonal-distance residuals, and — per established practice — exclusion of the optically distorted band at the contact line, with the fitted profile extrapolated to the baseline [4,5,9].

2.4 Error Sources in Optical Goniometry

Vuckovac et al. [9] quantified the dominant error: a ± 1 px error in baseline placement produces ≈ 0.5 – 1° of angle error for $\theta < 150^\circ$, growing steeply to $\approx 8^\circ$ as $\theta \rightarrow 180^\circ$; the error scales inversely with the drop's size in pixels, and seven expert users analysing the same image spread over $\sim 10^\circ$. Three design rules follow, all adopted here: maximise drop pixel size; determine the baseline algorithmically and to sub-pixel precision (on reflective substrates, from the drop-reflection junction — the approach of Krüss ADVANCE [13] and of the Dropometer [14]); and propagate baseline sensitivity into the reported uncertainty. Additional documented sources — stage tilt, autofocus scale drift, highlight clipping destroying the 50%-intensity edge, and the near- 90° contact-point localisation band [11] — are addressed in the capture subsystem (§3.3) and quality flags (§4.7).

2.5 Prior Art

Smartphone-based goniometry is established: Chen, Muros-Cobos and Amirfazli demonstrated research-grade accuracy on a phone [14]; low-cost hardware designs appear in HardwareX [15] and the educational literature [16]. Open-source analysis tools (ImageJ DropSnake/LB-ADSA [6,7], OpenDrop [17], pyDSA, Sessile.drop.analysis, DropPy [18]) document the algorithmic state of the art; machine-learning approaches [19,20] trade classical precision for robustness on messy images. The niche addressed here is the combination of: fully on-device native analysis, a multi-detector baseline hierarchy spanning matte through glossy/reflective substrates, ensemble fitting anchored by an authoritative ADSA fit, and — most distinctively — a locked-in, three-tier quantitative validation harness.

Chapter 3: Instrument and Software Architecture

3.1 Hardware

The optical bench follows the commercial goniometer layout at fabrication cost: a uniform LED back-light behind the drop, an X/Z sample stage (scissor lift for height, micrometer translation stage for lateral positioning), and a multi-axis smartphone mount on an acrylic base. The design was digitally prototyped as a fully-constrained SolidWorks assembly; after a feasibility analysis showed that sourcing low-volume precision components locally was prohibitively expensive, the project pivoted to a design-for-fabrication approach and professional vendor fabrication. The geometry produces the canonical measurement image: a dark drop silhouette on a bright, even field, resting on the darker substrate/stage band. Total hardware cost $\approx$ ₹18,000 against $\approx$ ₹15,00,000 for a commercial instrument.



Fig. H1: The fabricated instrument. Left: complete integrated setup — LED back-light, sample stage and smartphone mount on the acrylic base. Right: the fabricated sample stage — scissor lift (Z) with micrometer translation stage (X).

3.2 Application Architecture

The application is written entirely in Dart on Flutter. The analysis backend is native Dart with no OpenCV dependency in the measurement path, which keeps the pipeline portable, testable on the development machine, and identical across devices. Analysis of a full-resolution image completes in $\approx 2\text{--}5$ s on a mid-range phone. The UI provides gallery import, an in-app metrology camera, an optional region-of-interest selector (the standard ADSA workflow for excluding background contamination), an annotated result overlay, a full method/uncertainty breakdown, measurement-quality warnings, and CSV export of every numeric output for laboratory records.

3.3 Metrology-Aware Capture Subsystem

Because capture geometry, not the sensor, limits accuracy (§2.4), the in-app camera enforces measurement discipline:

Attitude readout. The accelerometer drives a live badge showing *roll* (rotation about the optical axis — baseline leveling; green at $\leq 0.5^\circ$) and *pitch* (optical-axis tilt; ISO 19403-2 recommends the

camera look 0–4° downward so the contact points and reflection are visible [12]). A roll-rotated horizon line is drawn over the baseline guide, and the shutter ring turns green only when both are within tolerance.

Focus and exposure control. Tap-to-focus locks the focus (autofocus hunting changes magnification mid-session); exposure is biased -1 EV by default so the bright field stays below clipping and the drop stays solidly dark, and can be locked once set. Flash and HDR are disabled.

Framing aids. A draggable baseline guide, rule-of-thirds grid and centre line; zoom is provided so the operator can stand back and zoom — a poor-man's telecentricity that reduces perspective error compared to shooting close.

Post-capture quality control. Every captured frame is immediately analysed for highlight clipping ($>2\%$ saturated pixels indicates blooming, which destroys the sub-pixel edge) and for silhouette contrast; a failing frame triggers a retake prompt before it can enter analysis.

Chapter 4: The Image-Analysis Pipeline

4.1 Silhouette Extraction with Sub-Pixel Edges

Back-lit drops defeat classical gradient edge detection: the dark drop merges into the dark substrate (no edge exists between them), while the interior refraction window creates strong spurious edges. The extractor instead works from the one reliable contrast — bright background versus dark object:

(i) Global Otsu threshold [21] separates the bright field from dark objects. **(ii) Baseline detection** follows (§4.2), yielding a possibly tilted line. **(iii) Connected-component selection** above the baseline isolates the drop from the substrate it visually merges with, scoring components by area, baseline support and centrality; a top-of-frame “vignette band” penalty is applied only to components whose mass hugs the top of the frame, so wide needle-topped drops are retained. **(iv) Row-scan of the outer silhouette** takes, per row, the outermost component pixels — inherently immune to interior refraction edges — continuing to the deepest per-column baseline so that stage tilt does not amputate the downhill contact region; a flank is accepted only when the pixels just outside it are bright background, which rejects baseline-crossing artefacts. **(v) Sub-pixel refinement:** each edge is refined to the 50%-coverage intensity crossing between the local background and object levels — the exact boundary location for an anti-aliased edge — and the same vertical crossing refines every baseline-fit point, removing the ≈ 0.5 px systematic bias of integer-row baselines (worth $\approx 0.5^\circ$ on small drops, since $d\theta \approx \sigma_{\text{baseline}}/a$).

All contour points are then rotated into the baseline-aligned frame (a true rotation, not a shear), in which every subsequent fit operates; left and right angles are computed independently and tilt-corrected, the standard practice across instruments and open tools [6,13,18].

4.2 Baseline Detection Hierarchy

Because baseline error dominates the goniometry error budget [9], the pipeline stacks four detectors, each matched to a physical substrate regime, with strict arbitration:

(a) Matte band clustering. On a matte dark stage, the per-column first-dark rows cluster at the substrate top; a robust two-pass line fit with inlier re-selection yields the baseline and its tilt. A *flank-support* requirement (inliers must span both outer flanks) distinguishes a genuine full-width substrate band from the drop's own upper edge.

(b) Specular mirror symmetry. On mirror stages (silicon, polished metal) there is no dark band; the drop and its reflection form a blob symmetric about the surface plane with a narrow waist at the contact line ($\theta > 90^\circ$). The baseline is located as the sub-pixel symmetry axis of the row-darkness profile, with a strict two-lobes-plus-waist requirement so the detector abstains on matte scenes — the reflection is treated as signal, the founding idea of DropSnake [6,7].

(c) Glossy-surface step detection. Mid-gray glossy substrates (e.g. Teflon viewed with a slightly downward camera) are *brighter than the Otsu threshold* and hence invisible to (a). Their surface appears on the drop-flanking columns as a bright→mid-gray transition that can be a very gradual ramp (≈ 1 grey level/row); it is localised per column as the sub-pixel crossing of the midpoint

between the two intensity plateaus — precise regardless of ramp width — using the *first* sustained step from the top (a stage with a dark front edge otherwise dominates the response). Only columns containing no dark object may vote, which automatically excludes the drop, the needle, and any burned-in annotations.

(d) Drop/reflection junction refinement. The flank horizon found by (c) is the substrate's far edge; the true contact line lies below it, where the drop meets its reflection. Following the “contour discontinuity” practice of Krüss ADVANCE [13], the Dropometer's slope-sign flip [14] and patent DE10214439A1 [22], each flank's outermost-x trace is scanned for its corner — a V-vertex for $\theta < 90^\circ$, a waist vertex for $\theta > 90^\circ$ — by two-segment line fitting with sub-pixel intersection; the line through the two corners provides both the baseline and its tilt. Fallbacks cover the near- 90° regime (Chen et al.'s one-third-of-vertical-run rule [14]) and a mirror-symmetry axis on the trace; a width-bulge test arbitrates against the matte line, exploiting the fact that at a genuine junction the blob narrows below the contact whereas at a true matte contact the width maximum sits on the matte line itself.

Arbitration rules were each validated on real failure cases: a flank-supported matte line is only overridden by a corner-confirmed junction; an unconfirmed step horizon loses to any matte line below it; and equator ambiguity (a circle is locally mirror-symmetric about its widest row) forbids the weaker fallbacks from overruling a supported matte line. Every result records which detector fired (`baseline_method`) for auditability.

4.3 Needle-in-Drop Handling

A dosing needle entering the apex is detected on the traced silhouette as a sustained narrow run at the top whose median width is far below the drop's maximum width; those rows are trimmed to the shoulder where the true drop widens. Component selection retains needle-topped drops (§4.1-iii), and the literature-standard alternative — masking $\pm 2\times$ needle width and fitting each side independently [23] — is recorded as future work for heavily distorted cases.

4.4 Geometric Fitting Methods

Circle. Algebraic least-squares circle with a geometric R^2 ($1 - \Sigma(d_i - r)^2 / \Sigma\|p - \bar{p}\|^2$); θ from the circle–baseline intersection. The validity gate requires the circle's *top* to rise above the baseline — the centre itself correctly lies below the baseline for hydrophilic caps, a condition that in an earlier version silently disabled the method for all $\theta < 90^\circ$.

Ellipse. Numerically-stable direct conic fitting after Halir–Flusser [24], with the eigenvector selected through a full 3×3 characteristic-polynomial eigensolver (plain power iteration selects the wrong conic branch), and tangents computed analytically at the baseline intersections.

Rotated-frame polynomial tangent. Fitting $y(x)$ or $x(y)$ in the image frame folds wherever the contour turns past the chosen axis inside the fit window — an unavoidable fold for steep flanks that biased the tangent method by -5° at 145° . The window is therefore rotated into the local tangent frame estimated by weighted PCA of the near-contact points, in which the profile is single-valued for any contact angle; a low-order and a higher-order fit must agree for the result to be accepted.

Radial weighting scales with the observed contact span, addressing the documented order/pixel-count sensitivity of polynomial methods [6].

4.5 Axisymmetric Drop-Shape Analysis (ADSA)

The reference method integrates equation (2) from the apex with a fixed-step RK4 scheme (local error $\approx 10^{-9}$, far below pixel noise), and fits the profile to the extracted contour by minimising *orthogonal* distances — vertical residuals bias steep-profile regions [4,9] — over the parameter vector (β, b, x_0, z_0) , using a multi-start Nelder–Mead simplex [25] over physically spaced Bond seeds with robust trimmed residuals. Three metrological details:

Contact-zone exclusion. The band immediately above the contact line is optically distorted (blur, meniscus, residual reflection); following ADSA-NA/selected-plane practice [4,5,9] those pixels are excluded from the fit and θ is read by *extrapolating* the fitted profile to the true baseline. On this instrument the distortion band was measured at ≈ 2 px; a proportional band (5% of drop height) was tested and found strictly worse on both synthetic truth and laboratory references — the exclusion is therefore fixed, not proportional.

Degeneracy gates. A near-flat, huge-apex-radius solution can achieve a spuriously small b-normalised residual and a high R^2 on wide drops; a physical gate rejects any fitted θ below $0.6 \times$ the spherical-cap bound $2 \cdot \text{atan}(h/a)$. Separately, fitted $\beta < 0.05$ sets a `low_bond_degenerate` flag: the angle remains valid but the Young–Laplace fit carries no information beyond the circle fit [8].

Authoritative anchoring. When the ADSA fit achieves near-perfect geometric quality it anchors the ensemble rather than being averaged against geometrically cruder methods (§4.6).

4.6 Ensemble Combination and Validity Gating

Each method first passes a validity gate (fit quality, contact consistency, plausibility); survivors are combined by weights built from fit quality, contact confidence and symmetry, moderated by a Tukey penalty on disagreement with the robust median and a leave-one-out consistency check. An authoritative ADSA fit bypasses the outlier penalty by design. If all models fail, a local-tangent rescue path reports a low-confidence result rather than silence. The final report contains the ensemble angle, per-method angles, left/right angles, and every method's acceptance/rejection reason.

4.7 Uncertainty Budget and Quality Flags

Each measurement carries a combined standard uncertainty from five terms added in quadrature (Table 5): a moving-block bootstrap over the contour (blocks $\approx \sqrt{n}$ preserve the spatial correlation of edge noise; an i.i.d. bootstrap would underestimate); inter-method disagreement (robust MAD); edge localisation from near-contact scatter; contact-confidence mapping; and — the term the literature identifies as dominant [9] — baseline-placement sensitivity, which for a circular cap is exact:

$$d\theta/dh = 1/(r \cdot \sin \theta) = 1/a \quad (\text{rad per pixel of baseline error}) \quad (3)$$

reproducing both the published $\approx 0.5^\circ/\text{px}$ plateau below 150° and the blow-up towards 180° ; σ_{baseline} is taken from the baseline fit's standard error, floored at a conservative sub-pixel value. ISO-style quality flags accompany the result: $\theta \geq 150^\circ$ (steeply growing baseline sensitivity), $\theta \leq 20^\circ$ (ambiguous tangent), stage tilt $> 2^\circ$, left–right asymmetry $> 4^\circ$ (non-axisymmetric drop or baseline error), and small drop (< 100 px half-width; error scales inversely with drop pixel size [9]).

Term	Estimator	Typical (this rig)
Contour sampling	moving-block bootstrap, 95% interval	$\pm 0.1\text{--}0.5^\circ$
Method disagreement	$1.4826 \times \text{MAD of valid methods}$	$\pm 0.2\text{--}1.5^\circ$
Edge localisation	near-contact residual spread / drop radius	$\pm 0.12\text{--}0.5^\circ$
Contact confidence	calibrated penalty map	$\pm 0.15\text{--}1.8^\circ$
Baseline placement	$\sigma_{\text{baseline}}/a$ (eq. 3)	$\pm 0.1\text{--}0.6^\circ$

Table 5: Uncertainty budget terms combined in quadrature.

Chapter 5: Validation and Results

Accuracy claims rest on three independent tiers, all encoded as automated regression tests with locked thresholds, so any change that degrades accuracy fails the build.

5.1 Tier 1 — Known-Truth Synthetic Harness

Laboratory references are themselves fits; only synthetic images provide exact truth. The harness renders anti-aliased spherical-cap silhouettes by 8×8 -supersampled coverage — placing the true edge exactly at the 50%-intensity crossing — over $60\text{--}145^\circ$, with stress variants adding Gaussian sensor noise, $\pm 4\text{--}5^\circ$ stage tilt, defocus blur, small drops ($R = 65$ px), and the contact-shadow wedge of §5.4.

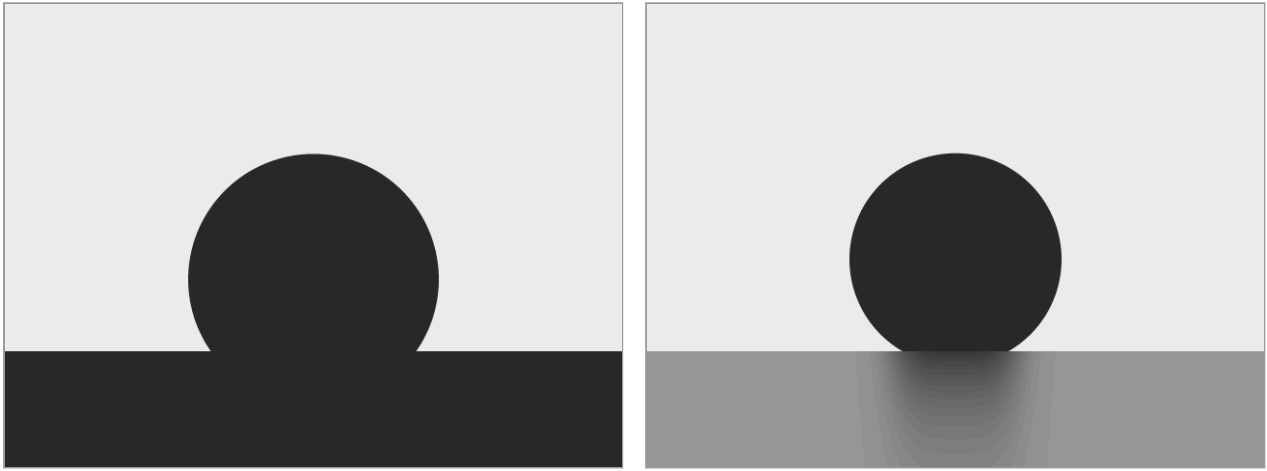


Fig. 1: Known-truth synthetics. Left: clean 125° cap. Right: 150° cap on a mid-gray substrate with a blocked-light contact-shadow wedge and outward penumbra.

θ true	Ensemble	error	ADSA	Circle	Polynomial	Ellipse
60	59.91	0.09	59.96	59.98	59.49	59.81
75	74.93	0.07	74.93	75.00	74.76	75.00
85	84.93	0.07	84.95	85.00	84.59	85.01
95	94.86	0.14	94.89	94.98	94.38	94.94
105	104.91	0.09	104.93	104.98	104.63	104.92
115	114.91	0.09	114.95	114.99	114.48	114.89
125	124.86	0.14	124.89	124.99	124.40	124.82
135	134.92	0.08	134.90	135.01	134.92	134.76
145	144.81	0.19	144.85	145.00	144.52	144.43

Table 1: Clean synthetic recovery (degrees). Ensemble MAE = 0.106° , maximum error 0.19° ; ADSA alone 0.083° .

Stress case	θ true	Ensemble	error
Gaussian noise $\sigma=4$	115	114.93	0.07

Gaussian noise $\sigma=4$	135	134.86	0.14
Gaussian noise $\sigma=8$	125	124.88	0.12
Stage tilt $+5^\circ$	120	119.92	0.08
Stage tilt -4°	130	129.91	0.09
Small drop $R=65$ px	115	114.97	0.03
Small drop $R=65$ px	140	139.68	0.32
Defocus blur $r=2$	125	124.90	0.10
Defocus blur $r=3$	140	139.79	0.21
Defocus blur $r=2$	75	74.90	0.10
Contact shadow (§5.4)	150	149.69	0.31
Contact shadow (§5.4)	135	134.97	0.03

Table 2: Stress recovery. MAE = **0.135°** , maximum error 0.32° . Notable: the $+5^\circ$ tilt case cost 1.76° before the tilted-baseline contour completion; it now costs 0.08° .

5.2 Tier 2 — Laboratory Reference (PFOTES Series)

Twelve drops on PFOTES-type fluorosilane-coated substrates ($\theta \approx 112\text{--}134^\circ$), captured on the fabricated rig, were referenced against LB-ADSA fits performed in the laboratory (ImageJ plugin [7]). One reference transcription error (132.4° recorded as 112.4°) was detected and corrected by re-reading the original fit screenshot — itself a demonstration of the audit value of keeping raw references.

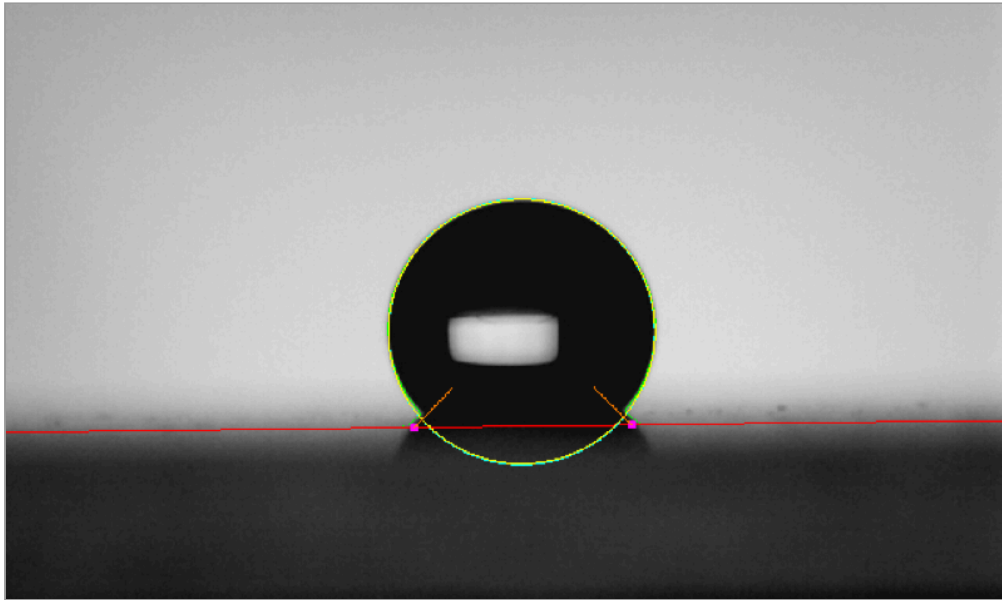


Fig. 2: Annotated analysis of a PFOTES drop: extracted contour (green), baseline (red), contact points (magenta) and fitted model overlays.

Sample	Reference LB-ADSA ($^\circ$)	This work ($^\circ$)	error ($^\circ$)
C 1.5% #1 coat 5a	112.07	111.12	0.95
C 1.5% #1 coat 5b	116.79	117.00	0.21

C 1.5% #1 coat 6	119.60	117.72	1.89
C 1.5% #2 coat 5	132.44	131.35	1.09
C 1.5% #2 coat 6	131.29	130.09	1.20
C 3% #1 coat 5	124.53	122.03	2.50
C 3% #1 coat 6a	131.62	129.62	2.00
C 3% #1 coat 6b	129.06	129.89	0.83
C 3% #2 coat 5a	114.54	114.52	0.02
C 3% #2 coat 5b	126.34	122.98	3.35
C 3% #2 coat 6a	133.68	134.55	0.86
C 3% #2 coat 6b	132.71	128.93	3.78

Table 3: PFOTES validation. MAE = 1.56° (1.2–1.3% relative), maximum 3.78°, ADSA converged on 12/12. The residuals are zero-mean scatter at the level of the reference method's own operator variability [9]; further tuning against 12 samples would constitute over-fitting.

5.3 Tier 3 — External-Dataset Generalisation (DropletLab)

Generalisation was tested on the published DropletLab Teflon dataset [26] — an adversarial set for this pipeline: dispensing needles in every frame, mid-gray glossy substrates viewed with a downward camera (so the visible flank horizon is the substrate's far edge, not the contact line), faint partial reflections, low resolution, and burned-in reference annotations. Sixteen reference angles were transcribed from the dataset's own Dropometer fits. The initial result (MAE $\approx 21^\circ$) was root-caused to the baseline regime described in §4.2(c–d); implementing the step + junction hierarchy, the component-selector correction and the ADSA degeneracy gate reduced the error to **7.3° MAE** — with the primary-rig validation (Tables 1–3) bit-for-bit unchanged.

Liquid group	n	MAE (°)	Median (°)	Max (°)
DI water	7	7.4	6.3	18.6
Ethylene glycol	4	2.1	1.8	5.6
Glycerol	5	11.4	12.5	20.7
Overall	16	7.3	—	—

Table 4: DropletLab external validation. The references are themselves polynomial fits on needle-distorted drops with left–right asymmetries up to 16° , so this figure bounds the combined disagreement of both methods, not this pipeline's error alone.

5.4 Case Study — the Contact-Shadow Baseline Hazard

A hazard identified early in the project (and common on this rig at high angles) is the dark wedge formed under the drop's overhang between the contact points, where blocked back-light merges the drop, its shadow and the substrate. A naive detector places the baseline at the bottom of the shadow, inflating the apparent contact width and corrupting θ . Fig. 3 shows the pipeline's behaviour on a real rig capture: the baseline locks to the substrate horizon and the contact points sit at the wedge apexes; the drop measures 134.95° with left/right agreement of 0.01° and independent-

method agreement (circle 135.6° , ADSA 135.9° , ellipse 132.4°), self-consistent with the cap relation $a/R = \sin \theta$. The synthetic shadow-wedge cases of Table 2 turn this from a verified example into a proven, regression-guarded capability: 150° recovered to 0.31° and 135° to 0.03° in the presence of the wedge and penumbra.

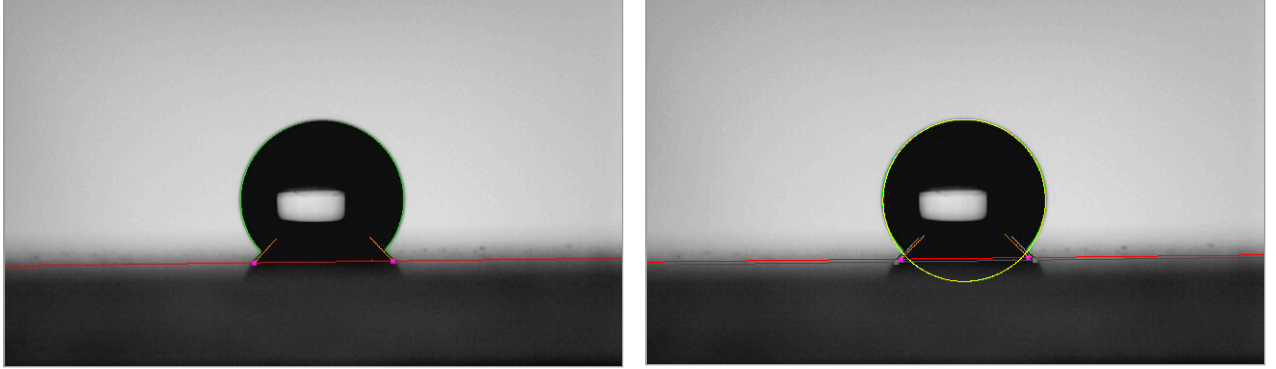


Fig. 3: Left: rig capture with contact-shadow wedge (reference tool's annotations burned in). Right: this pipeline's result — baseline (red) on the substrate horizon, contacts (magenta) at the wedge apexes, $\theta = 134.95^\circ$ with 0.01° left-right symmetry.

5.5 Known Limitations

Shadow-skirt regime. On the DropletLab glycerol images a soft dark halo between the drop and the substrate's front edge widens *monotonically* — there is no wedge apex, waist or corner for any trace-geometry detector to find — and the baseline lands low, costing $12\text{--}20^\circ$. The literature solution is a full mirrored-model image-energy fit (mirror the fitted profile below a candidate contact line and optimise its gradient energy against the image, per Stalder [6,7]) or learned segmentation [19,20]; this is scoped as future work. **Small drops.** Errors scale inversely with drop pixel size [9]; the application warns below 100 px contact half-width. $\theta \rightarrow 180^\circ$. Baseline sensitivity grows without bound (eq. 3); results above 150° are flagged, consistent with published recommendations to prefer force-based methods in that regime [9].

Chapter 6: Future Work

- 1. Mirrored-model baseline for the shadow-skirt regime.** Implement Stalder's principle directly: extend the fitted Young–Laplace profile below a candidate contact line by mirror symmetry and optimise the line's offset and tilt against the image's directional gradient energy [6,7]. This addresses the one remaining baseline regime (§5.5) with a 1–2-parameter continuous search rather than segmentation.
- 2. Needle-side independent fitting.** For heavily needle-distorted drops, adopt the circumcircle-plus-residual-polynomial per-side scheme of Albert et al. ($\leq 0.6^\circ$ reported accuracy, robust to polynomial order) [23].
- 3. Video and dynamic angles.** Short-clip capture with per-frame analysis over a stability plateau (mean $\pm$ SD reported, first frames discarded) — the established practice; $\sqrt{N}$ averaging does not hold because vibration is temporally correlated. Advancing/receding measurement follows with a tilt stage.
- 4. Scale calibration.** An in-frame reference (needle of known gauge, or stage fiducial) to convert pixels to millimetres, enabling absolute drop volume, physical Bond number and surface-tension estimation from the already-implemented ADSA fit in its identifiable regime [8].
- 5. Erf-model edge refinement.** Sub-pixel edges by fitting an error-function profile along edge normals — the back-lit edge is literally an erf — with the fitted blur width as a free focus-quality metric. Measured gain on this rig's synthetic tests was negligible, so this is deferred until heavy asymmetric blur becomes relevant.
- 6. On-device model fallback.** A small segmentation network ($\approx 1\text{--}3$ M parameters, quantised) for drop/reflection/needle/substrate separation on messy field images, used only when the classical hierarchy abstains [19,20].

Chapter 7: Summary and Conclusion

This project delivered a complete, validated, low-cost contact-angle metrology system: a fabricated back-lit optical bench ($\approx ₹18,000$; $>95\%$ below commercial cost) and a fully native mobile application implementing the reference physics (Bashforth–Adams ADSA with orthogonal residuals, contact-zone exclusion and degeneracy gating), robust geometry extraction (sub-pixel silhouette and baseline, a four-detector baseline hierarchy spanning matte, specular, glossy and reflective substrates, needle trimming), ensemble fitting with validity gating, and metrologically honest reporting — propagated uncertainty with the literature-dominant baseline-sensitivity term, per-method transparency, and ISO-style quality warnings.

Accuracy is demonstrated rather than asserted: 0.106° mean absolute error against exact synthetic truth (0.135° under noise, tilt, blur, small-drop and shadow stress), 1.56° against laboratory LB-ADSA references on the target coating system — at the reference method's own scatter — and 7.3° on a deliberately adversarial external dataset, a 65% error reduction achieved through literature-driven baseline research with zero regression on the primary rig. All thresholds are locked into an automated test suite, making the accuracy claims reproducible and protected against regression. The launcher icon itself (Fig. 4) renders the measurement this instrument performs: a sessile cap on its baseline with the tangent-and-arc mark.

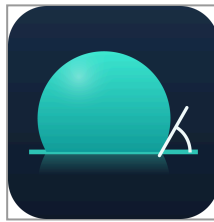


Fig. 4: Application icon, rendered programmatically from the measurement geometry (120° spherical cap, baseline, tangent and θ arc, faint reflection).

The combination of accessible hardware, on-device physics-based analysis, and a regression-guarded validation harness fulfils the project's goal: laboratory-defensible static contact-angle measurement, accessible at a small fraction of the cost of commercial instrumentation.

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